



2024

I20

BOOKLET NO.

300201

Forest Services
Geology

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

SEAL

P.T.O.

**SECTION – A**

Q1. Write short notes on **any five** from seven. **(8×5=40)**

- (a) Explain in brief Geographical Information System and its applications.
- (b) Give the primary classification of faults and explain it with neat diagrams.
- (c) Give brief account on Earthquake resistance structures.
- (d) Explain the Petrography and Petrogenesis of Granite.
- (e) Write detail account on retrograde metamorphism.
- (f) Discuss in detail the evolutionary trends in Huminidae.
- (g) Describe briefly the Cambrian-Precambrian boundary problem.

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- Q2.** (a) Explain in detail the geomorphic cycle concept with neat diagrams. **15**
(b) Describe in detail the types of folds with neat diagrams. **15**
(c) Explain different aerial photointerpretation elements with their merits and limitations. **10**

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- Q3.** (a) Give detail account of petrogenetic significance of the textures of igneous rocks. **15**
(b) Define metamorphism. Give types and agents of metamorphism. **15**
(c) Define Tunnel. Describe the geological investigations carried out for selection of tunnel site. **10**

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- Q4.** (a) Discuss in detail morphology, geological history and evolutionary trend in Cephalopoda. **15**
(b) Explain the term "Stratigraphy". Describe the equivalence of litho, bio and chronostratigraphic units. **15**
(c) Discuss different geomorphic theories of Landform development. **10**
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**SECTION - B**

Q5. Write short notes on **any five** from seven. **(8×5=40)**

- (a) Describe the inner planets of solar system.
- (b) What are meteorites ? Add a note on siderolites.
- (c) Explain any four types of crystal defects.
- (d) Write a brief account on manganese module.
- (e) Discuss in short about catagenetic stage of diagenetic process.
- (f) Salt water intrusion is a major problem on Indian coast. Justify.
- (g) Briefly describe the causes of floods.

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- Q6.** (a) Draw the internal structure of Earth and describe the composition of each layer. **15**
- (b) What are twinned crystal ? Discuss any four types of twins. **15**
- (c) How are pegmatites form ? Give their significance to granite related ore forming process. **10**

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- Q7.** (a) Describe various components of limestone. Discuss the Folk's classification of limestone. **15**
- (b) Draw and describe various types of aquifers according to geological formation. **15**
- (c) What is radioactive waste ? Explain the various methods for disposal of radioactive waste. **10**

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- Q8.** (a) Describe the genetic classification of ore deposits. **15**
- (b) What are earthquake ? Explain primary, secondary and tertiary hazards associated with earthquake. **15**
- (c) What are cleavage plane ? Give various types of cleavage exhibited by minerals. **10**
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